

Project Initialization and Planning Phase

Date	02 July 2026
Team ID	xxxxxx
Project Title	Smart Lender – Loan Eligibility Prediction Using Machine Learning
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The **Smart Lender** project is a machine learning-based web application developed to predict the loan eligibility of applicants. The system analyzes applicant information such as gender, marital status, education, self-employed status, applicant income, co-applicant income, loan amount, loan amount term, credit history, property area, and dependents to determine whether a loan should be approved or rejected. Using the trained **XGBoost** machine learning model integrated with a **Flask** web application and deployed on **Render**, the solution provides quick, accurate, and reliable loan eligibility predictions, reducing manual effort and improving the loan evaluation process.

Project Overview	
Objective	To develop a machine learning-based loan eligibility prediction system that accurately determines whether an applicant is eligible for a loan based on financial and personal details using classification algorithms and a user-friendly Flask web application.
Scope	The project focuses on predicting loan eligibility using historical applicant data and machine learning algorithms. It includes data preprocessing, exploratory data analysis, model training using Decision Tree, Random Forest, K-Nearest Neighbors (KNN), and XGBoost , model evaluation, prediction, and deployment through a Flask web application hosted on the Render cloud platform. The system is intended for educational purposes and can be extended for real-world banking applications.
Problem Statement	
Description	Banks receive numerous loan applications every day. Manual verification of applicant details is time-consuming, error-prone, and may lead to inconsistent decisions. An intelligent automated system is required to assist financial institutions in predicting loan eligibility quickly, accurately, and consistently.
Impact	The Smart Lender system reduces loan processing time, improves prediction accuracy, minimizes manual effort, supports faster decision-making, and helps financial institutions make more reliable loan approval decisions while improving customer experience.

Proposed Solution	
Approach	The system collects applicant details through a Flask-based web interface , preprocesses the input data by handling missing values, encoding categorical features, and scaling numerical values. The processed data is passed to the trained XGBoost machine learning model, which predicts whether the loan application should be Approved or Rejected . The prediction result is displayed instantly through the deployed Flask web application hosted on Render .

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU
Memory	RAM specifications	8 GB RAM
Storage	Disk space for project files, datasets, and models	10 GB Free Disk Space
Software		
Frameworks	Python Web Framework	Flask
Libraries	Machine Learning and Data Analysis Libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, XGBoost, Joblib
Development Environment	IDE / Code Editor	Visual Studio Code, Jupyter Notebook
Data		
Data	Loan Eligibility Dataset	CSV dataset containing applicant details such as gender, marital status, education, income, loan amount, loan term, credit history, property area, and loan status used for training and testing the machine learning models.